JTMK Project Report:
Campus Network Simulation using Cisco Packet Tracer

Project Title:

Campus Network Simulation using Cisco Packet Tracer

Team Members:

• Muhammad Hail bin Mohd Ariff (Leader)
• Muhammad Alif Najmi Bin Sahri
• Muhammad Nafiz bin Mohamed Nazarudin
• Muhammad Amirul bin Norizan

Introduction:

The Campus Network Simulation project aims to replicate the structure and function of a secure campus-wide network using Cisco Packet Tracer. It simulates various networking protocols and configurations to help students understand real-world network design and implementation.

Objectives:

1. To design a simulated campus network using Cisco Packet Tracer.
2. To implement VLANs, DHCP, DNS, and NAT configurations.
3. To simulate firewall and access control mechanisms.
4. To provide internet access through simulated routing protocols.

Tools and Technologies Used:

• Software: Cisco Packet Tracer
• Protocols: VLAN, DHCP, DNS, NAT, RIP, ACL
• Devices: Routers, Switches, PCs, Servers
• Configuration: CLI and GUI within Cisco Packet Tracer

Features:

• VLAN segmentation for department separation.
• Dynamic IP assignment via DHCP server.
• Simulated DNS for domain resolution.
• Access control using standard and extended ACLs.
• NAT setup to allow private network internet access.

Implementation Process:

1. Planning: Identified departments and created network topology.
2. Device Setup: Configured routers, switches, and end devices.
3. Network Configuration: Assigned IPs, configured DHCP, DNS, and routing.

4. Security Setup: Implemented ACLs and firewall rules.
5. Testing: Verified connectivity and security protocols across VLANs.

Results:

- Successful simulation of a full campus network with 5 VLANs.
- All departments accessed the internet with NAT.
- ACLs effectively restricted inter-VLAN traffic as per design.
- DNS and DHCP functionalities were verified through testing.

Challenges:

- Configuring accurate ACL rules without blocking essential services.
- Maintaining proper routing tables in a multi-router environment.
- Synchronizing DNS resolution with dynamic IP addressing.

Future Enhancements:

- Integrate wireless access points for mobile device simulation.
- Add redundancy via HSRP or VRRP for critical routers.
- Simulate VoIP and video conferencing within the network.

Conclusion:

This project provided hands-on experience in configuring a secure and functional campus network. The simulation enhanced our knowledge of network protocols, security implementations, and practical troubleshooting techniques in a complex network scenario.

References:

1. Cisco Packet Tracer Help - https://www.netacad.com/courses/packet-tracer
2. VLAN Configuration Guide - https://www.cisco.com/c/en/us/td/docs/
3. DHCP and DNS Setup - https://www.geeksforgeeks.org/
4. ACL Tutorials - https://www.practicalnetworking.net/
5. NAT Implementation - https://www.cisco.com/c/en/us/support/docs/ip/network-address-translation-nat/